

Problem 26.12

There are two perpetuities-immediate. The first has level payments of p at the end of each year. The second is increasing such that the payments are $q, 2q, 3q, \dots$. Find the rate of interest that will make the difference in present value between these perpetuities

- (a) zero;
- (b) a maximum.

Solution

The present value of the level perpetuity is

$$PV_1 = \frac{p}{i}.$$

The present value of the increasing perpetuity is

$$PV_2 = q(Ia)_{\infty|i} = \frac{q}{id} = \frac{q}{i} + \frac{q}{i^2}.$$

Therefore, the difference in present values is

$$D(i) = PV_1 - PV_2 = \frac{p}{i} - \frac{q}{i} - \frac{q}{i^2} = \frac{p-q}{i} - \frac{q}{i^2}.$$

Part (a).

For the difference to be zero,

$$\frac{p-q}{i} - \frac{q}{i^2} = 0.$$

Multiplying by i^2 gives

$$i(p-q) - q = 0.$$

Thus,

$$\boxed{i = \frac{q}{p-q}}.$$

Part (b).

Differentiate $D(i)$ with respect to i :

$$D'(i) = -\frac{p-q}{i^2} + \frac{2q}{i^3}.$$

Set the derivative equal to zero:

$$-\frac{p-q}{i^2} + \frac{2q}{i^3} = 0.$$

Multiplying by i^3 gives

$$-i(p - q) + 2q = 0.$$

Therefore,

$$i = \frac{2q}{p - q}.$$

Moreover,

$$D''(i) = \frac{2(p - q)}{i^3} - \frac{6q}{i^4},$$

which is negative at $i = 2q/(p - q)$. Hence the difference is maximized at

$$\boxed{i = \frac{2q}{p - q}}.$$

These positive rates require $p > q$.